Scaling Saturation Genome Editing (SGE) at the Wellcome Sanger Institute

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Health and safety induction

Labs are category level 2

- Medium-risk biological agents, which can cause human or animal disease. This includes genetically modified organisms and certain pathogens.
- However, our cells are category level 1 and are non-hazardous

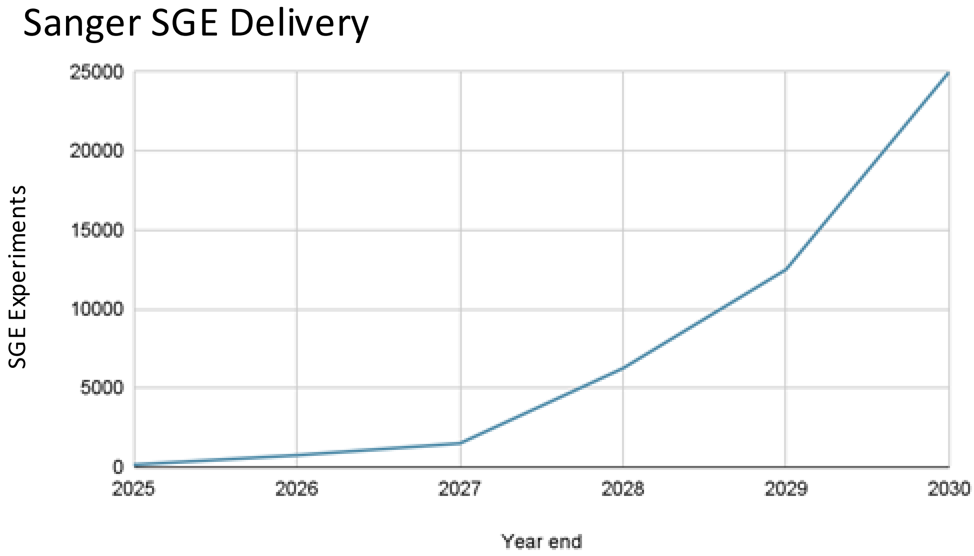
Lab coats and goggles/glasses must be worn at all times

Avoid touching equipment unless instructed to do so by the demonstrators

Wear gloves when instructed to do so by the demonstrators

If you need to leave the lab use the doorbell to allow re-entry

Aspiration of Scale

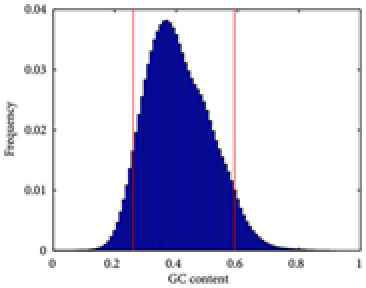


- Current SGE methods for ~20,000 genes across the human genome equal is to **~500,000 SGE experiments**
- SciOps pipeline aims to establish a pipeline to deliver **1000 genes in the next 5 years (~25000 SGE experiments)**

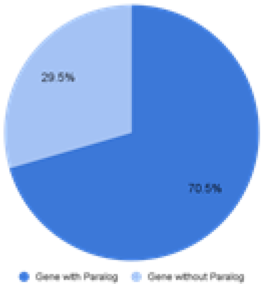
The challenges of scale

- Molecular biology:
  - Each genomic region has unique sequence characteristics (e.g. GC content, repetitive sequences, paralogous genes)

GC% of a sequence impacts molecular biology steps [1]



Paralog genes increases risk of off-target effects [2]

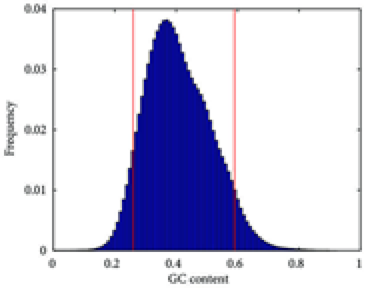


1. Duan,et al. Detection of False-Positive Deletions from the Database of Genomic Variants. BioMed Research International. 2019.
2. Zerbino et al. Ensembl 2018, Nucleic Acids Research, Volume 46, Issue D1, 4 January 2018

The challenges of scale

- Molecular biology:
  - Each genomic region has unique sequence characteristics (e.g. GC content, repetitive sequences, paralogous genes)
- Cellular biology:
  - SGE requires generating loss-of-function variants in cell lines
  - Highly manual, low-automation experimental workflow

GC% of a sequence impacts molecular biology steps [1]



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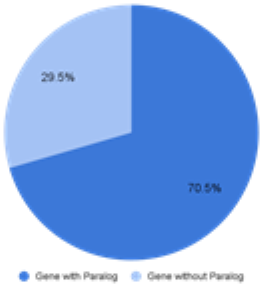

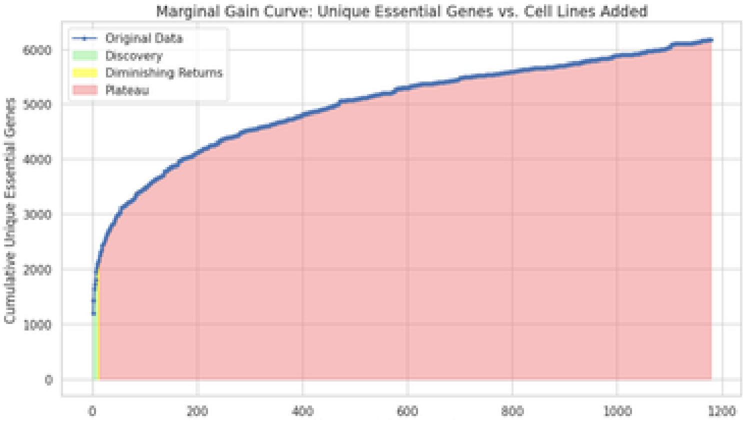

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We have established a cross functional pipeline to deliver SGE

Design	Constructio n	Screening	DNA Extraction	Sequencing Library Prep	Analysis
In silico bioinformatic design of SGE components	Molecular biology construction of plasmid vectors	High throughput culture of human cells for loss of function variants	Extraction of genomic DNA	Preparation of the specific genomic regions for sequencing	Analysis of data to identify loss of function variants

Informatics Data Solutions

Cellular Operations

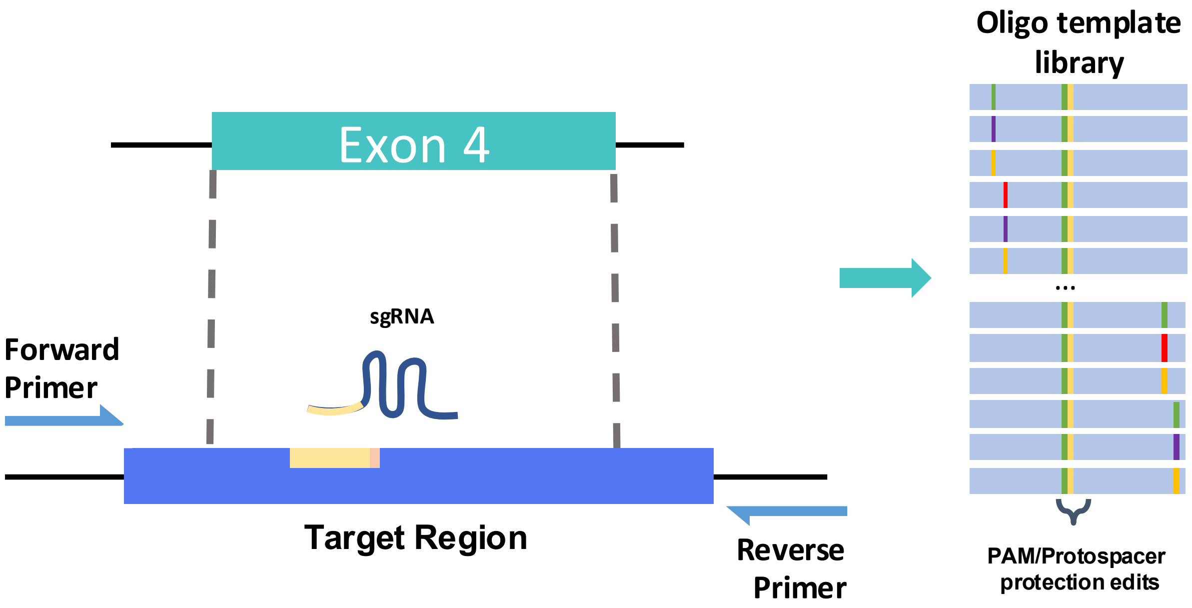
Sequencing Operations

Informatics Data Solutions

Demonstration

wellcome sanger institute

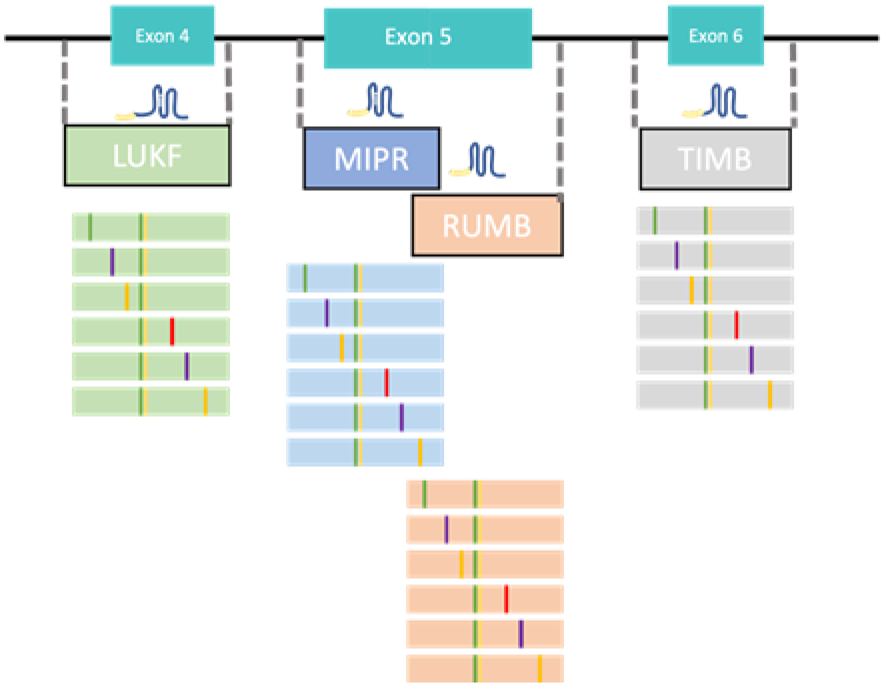
SGE in silico design



- Genomic region of interest is <300 nucleotides
- Defined by primers flanking the region for plasmid construction
- Includes an sgRNA targeting site
- Forms the basis for an oligonucleotide template library that contains every variant and clinically relevant mutations

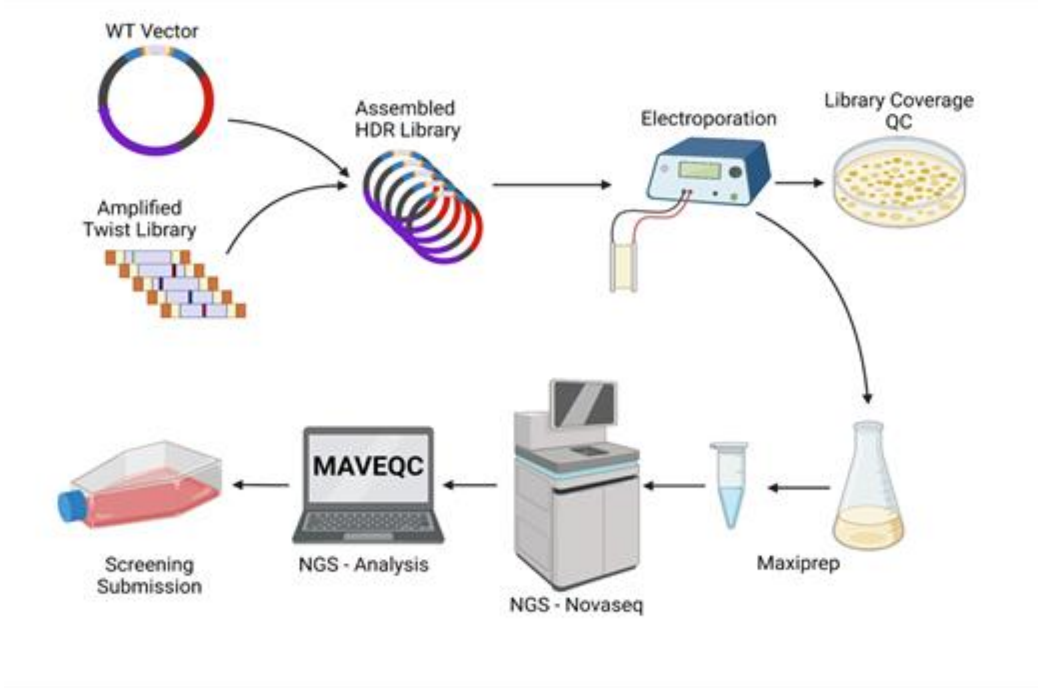
Slides produced by Leah Wells

SGE *in silico* design



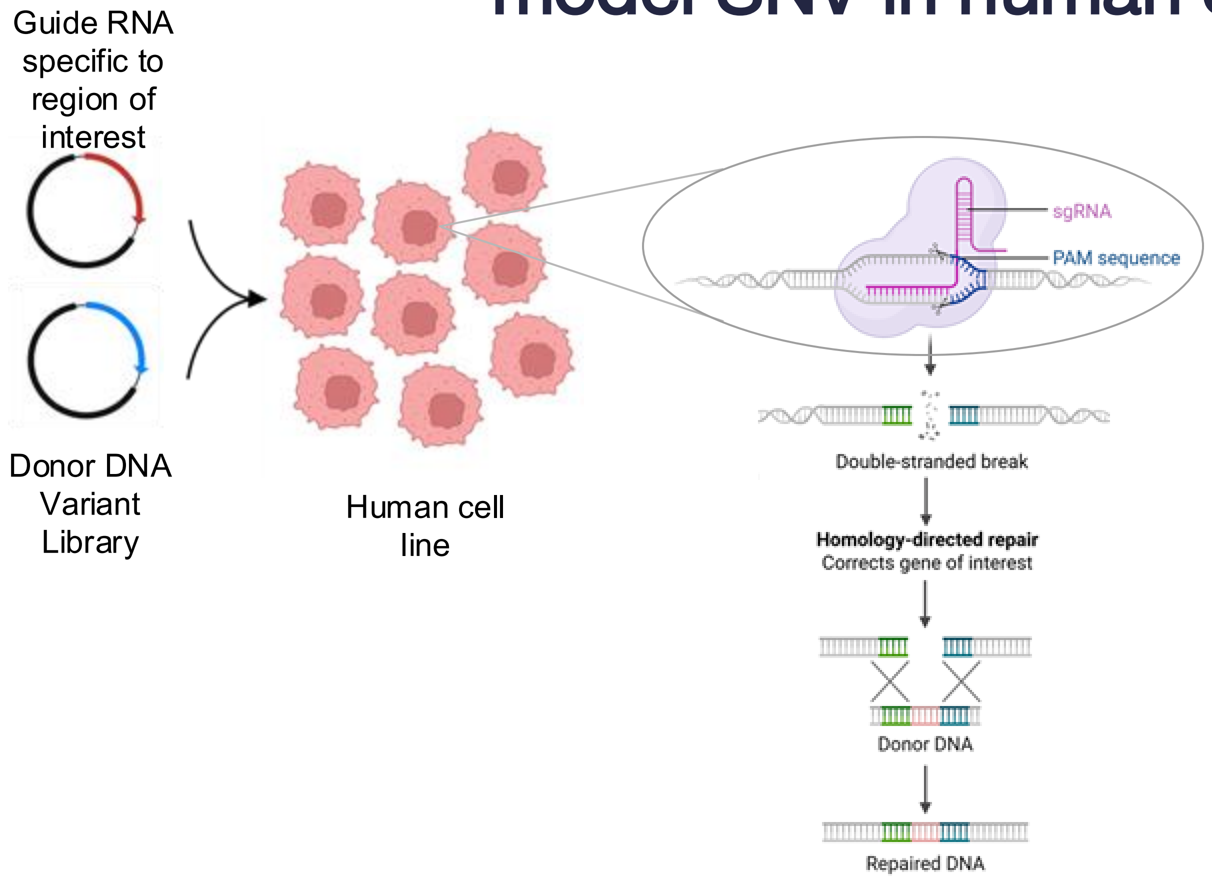
- Development of software tools to design all *in silico* components for any genomic region of interest
- Identification of optimal primer pairs to flank the target region
- Selection of optimal guide RNAs to maximise CRISPR/Cas9 efficiency
- Definition of fundamental design rules applicable across all sequence characteristics

Construction molecular biology steps to create donor DNA variant template



- Begin with the wild-type genomic sequence of the target region
- Introduce all potential SNVs into the region within a plasmid backbone
- Transform into bacteria for large-scale amplification
- Verify SNV representation in the final donor DNA Variant Library

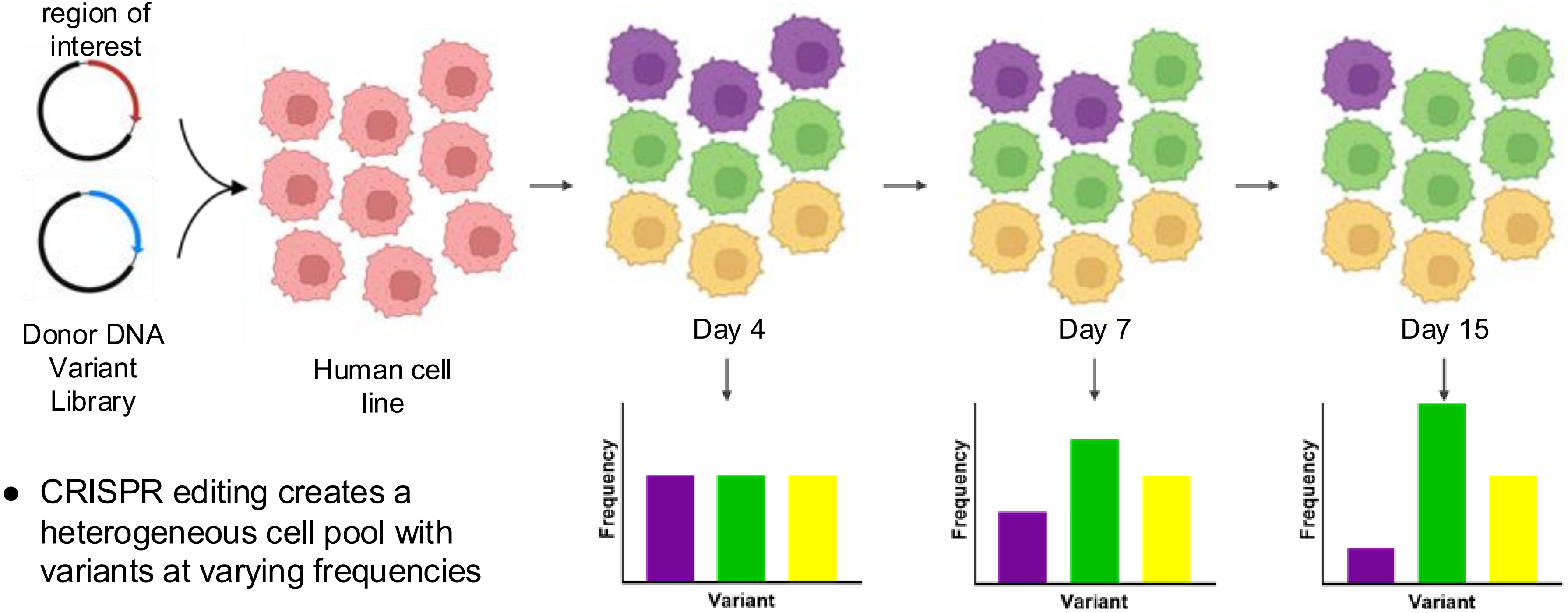
Using homology-directed repair to model SNV in human cells



- Utilise the cell’s homology-directed repair (HDR) mechanism and CRISPR/Cas9 to introduce SNVs via donor templates
- Generate a unique SNV in each individual cell
- Propagate cells; loss-of-function variants are eliminated in vitro

Slides produced by Ian Lobb

Using homology-directed repair to model SNV in human cells

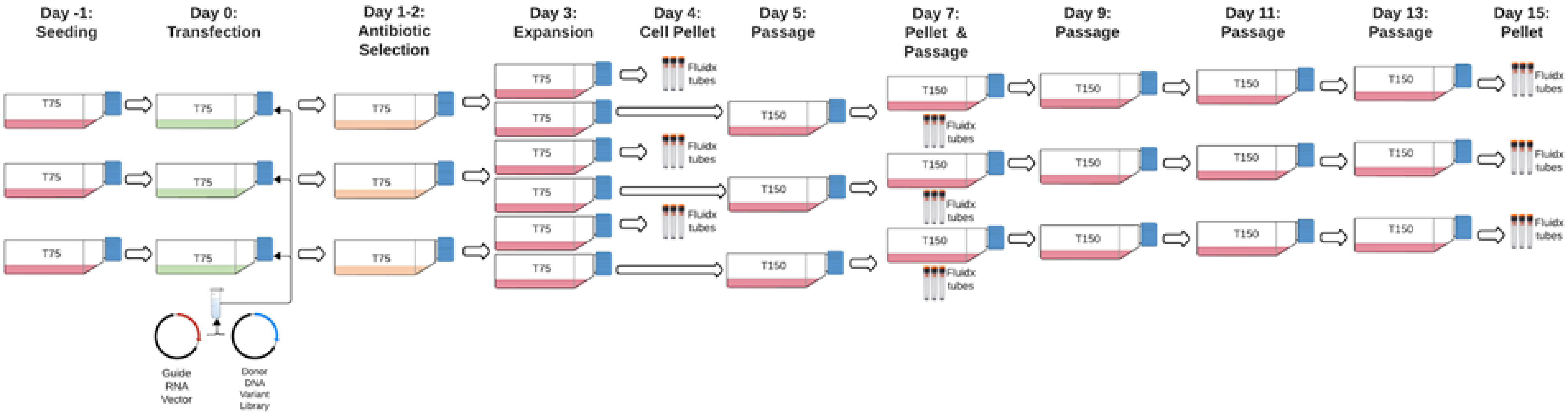


- CRISPR editing creates a heterogeneous cell pool with variants at varying frequencies
- Region of interest is quantified by amplicon PCR and sequencing

- Stable variant frequency → no effect on cell growth or survival
- Decreasing variant frequency → detrimental to growth or survival

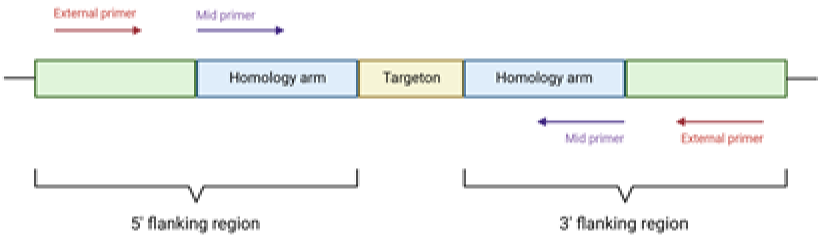
Slides produced by Ian Lobb

Screening cellular biology process to create SNV in the human genome

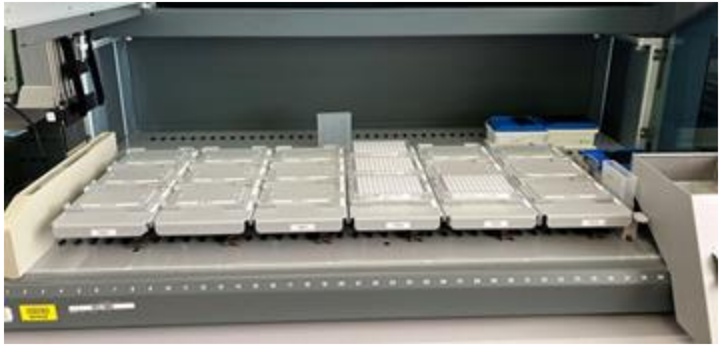


- This figure represents one saturation gene editing experiment covering ~250 nucleotides
- ~20 million cells transfected at day 0 to ensure hundreds of cells contain each SNV
- Puromycin resistance (via guide RNA vector) selects successfully transfected cells
- Cells cultured for 15–21 days, depending on gene depletion kinetics
- Samples frozen at days 4, 7, and 15 to capture depletion over time and identify early vs late depleting SNVs

Library Prep molecular biology process to identify presence of SNV absence with next generation sequencing



- Specific amplification of the targeton region where SGE was performed from genome
- Bespoke primers used for every SGE region
- Large scale automation to process hundreds of cell culture cell pellets



Tecan

Hamilton

Dragonfly